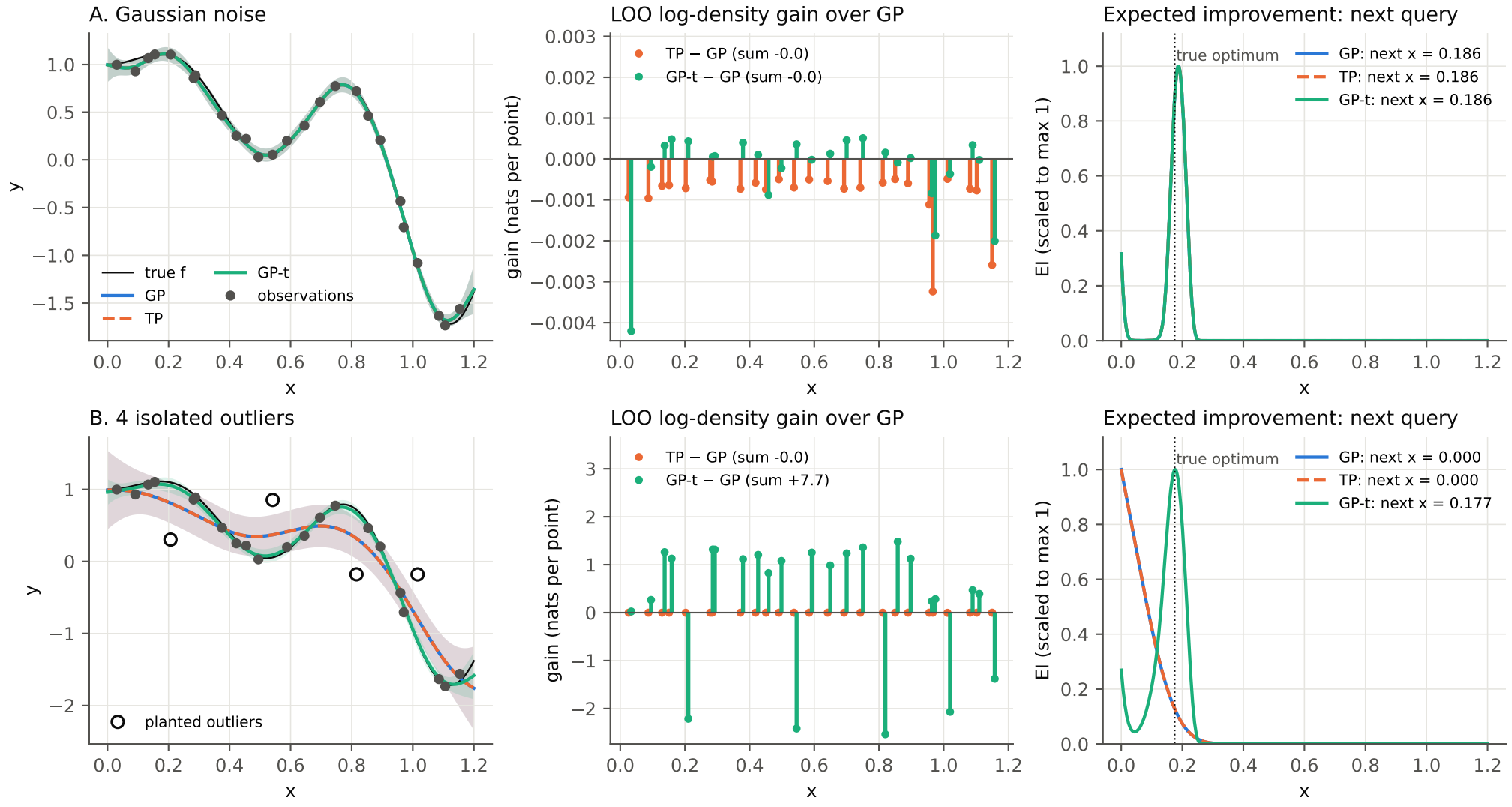

A. Gaussian noise
true f
GP
TP
GP-t
observations
x
y
LOO log-density gain over GP
TP − GP (sum -0.0)
GP-t − GP (sum -0.0)
gain (nats per point)
x
Expected improvement: next query
true optimum
GP: next x = 0.186
TP: next x = 0.186
GP-t: next x = 0.186
EI (scaled to max 1)
x
B. 4 isolated outliers
planted outliers
x
y
LOO log-density gain over GP
TP − GP (sum -0.0)
GP-t − GP (sum +7.7)
gain (nats per point)
x
Expected improvement: next query
true optimum
GP: next x = 0.000
TP: next x = 0.000
GP-t: next x = 0.177
EI (scaled to max 1)
x